H524 Assignment 1 - Answer Key

Introduction to Biostatistics and Data Visualization

Total Points: 103 points

Part A: Multiple Choice Questions (40 points, 4 points each)

Question 1: FEV1 Dataset Size

Correct Answer: b) 13 subjects

Question 2: Motor Vehicle Deaths Percentage

Correct Answer: c) 48%

Explanation: Motor vehicle accidents account for 48% of childhood injury deaths in the reference dataset.

Question 3: Variable Type - Gender

Correct Answer: c) Qualitative nominal

Explanation: Gender is a categorical variable with no natural ordering (nominal).

Question 4: Measure of Central Tendency for Skewed Data

Correct Answer: b) Median, because it's less affected by extreme values

Explanation: Median is preferred for highly skewed data as it's resistant to outliers.

Question 5: Box Plot Usefulness

Correct Answer: b) Display the five-number summary and outliers clearly

Explanation: Box plots show minimum, Q1, median, Q3, maximum, and outliers.

Question 6: SMR > 100 Interpretation

Correct Answer: b) Higher mortality than expected

Explanation: Standardized Mortality Ratio > 100 indicates observed mortality exceeds expected.

Question 7: R Function for Frequency Table

Correct Answer: b) `table()`

Explanation: The `table()` function creates frequency tables in R.

Question 8: Bar Charts vs Histograms

Correct Answer: a) Bar charts show categorical data with gaps between bars; histograms show continuous data with no gaps

Explanation: Bar charts are for categorical data, histograms for continuous data.

Question 9: Golden Rule for AI in Biostatistics

Correct Answer: b) AI should enhance understanding, not replace critical thinking

Explanation: AI is a tool to enhance learning, not replace it.

Question 10: NOT an AI Pitfall

Correct Answer: d) AI requires expensive commercial software

Explanation: Cost was not mentioned as an AI pitfall in the slides. The actual concerns are outdated methods, context confusion, and assumption violations.

Part B: Short Answer Questions (30 points)

Question 11: Data Types Classification (10 points, 2 points each)

a) FEV1 measurement in liters: - Type: Quantitative continuous - **Justification:** Measured on a continuous scale with infinite possible values between any two points; represents a measurable quantity with decimal precision.

b) Number of childhood injury deaths per state: - Type: Quantitative discrete - **Justification:** Countable whole numbers (counts); cannot have fractional values.

c) Cause of injury (motor vehicle, drowning, fire, etc.): - Type: Qualitative nominal - **Justification:** Categories with no inherent ordering or ranking.

d) Pain severity rating (1=mild, 5=severe): - Type: Qualitative ordinal (or discrete) - **Justification:** Categories with meaningful order (mild to severe), but intervals between values may not be equal.

e) Patient's primary language (English, Spanish, French, etc.): - Type: Qualitative nominal - **Justification:** Categories with no natural ordering.

Question 12: Geographic Health Data Analysis (10 points)

Map Data: - Oregon: 8.2% - Washington: 9.1% - California: 9.7% - Nevada: 10.8% - Idaho: 8.9%

a) Highest prevalence and geographic factors (3 points):

Nevada shows the highest diabetes prevalence (10.8%). Geographic factors that might contribute: - Desert climate with extreme temperatures may limit outdoor physical activity - Urban sprawl in Las Vegas area may promote sedentary lifestyles - Lower access to fresh produce in desert regions - Socioeconomic factors and demographics unique to Nevada

b) How visualization helps resource allocation (4 points):

Geographic visualization helps public health officials: - **Identify hotspots:** Quickly see which states/regions need immediate intervention - **Prioritize funding:** Allocate more resources to high-prevalence areas (Nevada) - **Target interventions:** Design region-specific programs based on spatial patterns - **Track progress:** Compare maps over time to evaluate intervention effectiveness - **Build coalition:** Show neighboring states (CA, NV) might benefit from coordinated efforts

c) Additional data layers for policy decisions (3 points):

Enhanced map layers could include: - **Demographics:** Age distribution, ethnicity, income levels - **Healthcare access:** Number of clinics, physicians per capita, insurance coverage - **Environmental factors:** Food desert locations, walkability scores, air quality - **Behavioral data:** Physical activity levels, dietary patterns, smoking rates - **Social determinants:** Education levels, unemployment rates, housing stability

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Question 13: Multi-Panel BMI Visualization Analysis (10 points)

Panel Description: - Panel 1: BMI distribution histogram with normal curve overlay - Panel 2: BMI comparison by gender using box plots - Panel 3: BMI category frequencies (Underweight, Normal, Overweight, Obese) - Panel 4: BMI vs systolic blood pressure scatter plot with regression line

a) Panel 1 vs Panel 2 comparison (3 points):

Panel 1 (Histogram): - Shows overall distribution shape and spread of BMI values - Reveals if data is normal, skewed, or multimodal - Good for understanding the overall population pattern - Shows frequency/density of values across BMI range

Panel 2 (Box plots by gender): - Compares BMI distributions between male and female groups - Shows median, quartiles, and outliers for each gender - Reveals gender differences in central tendency and variability - Better for comparing subgroups rather than overall distribution

b) BMI-Blood Pressure Relationship and Public Health Significance (4 points):

Relationship observed: - Positive correlation: As BMI increases, systolic blood pressure tends to increase - Linear trend indicated by regression line - Suggests obesity is associated with higher blood pressure

Public Health Significance: - **Cardiovascular risk:** Higher BMI linked to hypertension, a major CVD risk factor - **Intervention targets:** Weight reduction programs may help lower blood pressure - **Screening priorities:** High BMI individuals should be monitored for hypertension - **Prevention strategies:** Promoting healthy weight may reduce cardiovascular disease burden - **Health costs:** Addressing obesity could reduce hypertension-related healthcare expenses

c) Proposed fifth panel (3 points):

Suggested visualizations:

Option 1: Age vs BMI scatter plot (with color by gender) - **Health question:** How does the BMI-age relationship vary by gender? - **Value:** Identifies age groups at highest risk; guides age-specific interventions

Option 2: Geographic map of mean BMI by region - **Health question:** Are there geographic disparities in obesity prevalence? - **Value:** Targets resources to high-prevalence regions; identifies environmental factors

Option 3: BMI trends over time (line graph) - **Health question:** Is the obesity problem getting worse over time? - **Value:** Evaluates effectiveness of public health campaigns; forecasts future burden

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Part C: R Programming Exercises (30 points)

Question 14: R Data Analysis with FEV1 Dataset (15 points)

a) Mean FEV1 by gender (5 points):

```r
# Load the FEV1 data
source("week1_lecture_code.R")  # or create data directly

# Calculate mean FEV1 by gender
aggregate(fev1_liters ~ gender, data = fev1_data, FUN = mean)

# Alternative using tapply:
tapply(fev1_data$fev1_liters, fev1_data$gender, mean)
```

Expected Output:

```
  gender fev1_liters
```

```
1 Female    2.696667
2   Male    3.098571
```

Interpretation: Males have higher average FEV1 (3.10 L) compared to females (2.70 L), consistent with physiological differences in lung capacity.

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b) Summary statistics and quartile interpretation (5 points):

```r
# Summary statistics for FEV1
summary(fev1_data$fev1_liters)
```

Expected Output:

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  2.460   2.705   2.810   2.882   3.135   3.460
```

Interpretation of Quartiles: - 1st Quartile (Q1 = 2.705): 25% of subjects have FEV1 2.705 liters - **3rd Quartile (Q3 = 3.135):** 75% of subjects have FEV1 3.135 liters - **Interquartile Range (IQR = 3.135 - 2.705 = 0.43):** The middle 50% of values span 0.43 liters - **Interpretation:** Most lung function values cluster in a relatively narrow range (2.7-3.1 L), with 50% of subjects falling within this 0.43 L range.

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c) Frequency table for gender and percentage female (5 points):

```r
# Create frequency table for gender
gender_table <- table(fev1_data$gender)
gender_table

# Calculate proportions
prop.table(gender_table)

# Calculate percentage
prop.table(gender_table) * 100

# Or specifically for females:
female_count <- sum(fev1_data$gender == "Female")
total_count <- nrow(fev1_data)
female_percentage <- (female_count / total_count) * 100
female_percentage
```

Expected Output:

```
Female   Male
     6      7

Female      Male
0.4615385 0.5384615

Female     Male
 46.15    53.85
```

Calculation: 6 females out of 13 total subjects = 6/13 = 0.4615 = **46.2% female**

Question 15: Creating Geographic Health Visualizations (15 points)

a) Choropleth map code (8 points):

```r
# Load required libraries
library(ggplot2)
library(maps)
library(mapdata)
library(dplyr)

# Create state health data
state_health <- data.frame(
  state = c("oregon", "washington", "california", "nevada", "idaho"),
  diabetes_rate = c(8.2, 9.1, 9.7, 10.8, 8.9)
)

# Get map data
states_map <- map_data("state")

# Merge health data with map coordinates
map_data_merged <- left_join(states_map, state_health,
                              by = c("region" = "state"))

# Create choropleth map
ggplot(map_data_merged, aes(x = long, y = lat,
                             group = group, fill = diabetes_rate)) +
  geom_polygon(color = "white", size = 0.5) +
  scale_fill_gradient(low = "lightblue", high = "darkred",
                      name = "Diabetes\nRate (%)",
                      na.value = "grey90") +
  coord_quickmap() +
  theme_void() +
  labs(title = "Diabetes Prevalence by State",
       subtitle = "Pacific Northwest Region",
       caption = "Data: State Health Surveys") +
  theme(plot.title = element_text(size = 16, face = "bold"),
        legend.position = "right")

# Save the plot
ggsave("diabetes_choropleth.png", width = 10, height = 6, dpi = 300)
```

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b) Complementary visualization (4 points):

```r
# Create dot plot showing diabetes rates with state labels
ggplot(state_health, aes(x = diabetes_rate,
                          y = reorder(state, diabetes_rate))) +
  geom_point(size = 5, color = "darkred") +
  geom_segment(aes(x = 0, xend = diabetes_rate,
                   y = state, yend = state),
               color = "gray70", size = 1) +
  scale_x_continuous(limits = c(0, 12),
                     breaks = seq(0, 12, 2)) +
  labs(title = "Diabetes Prevalence by State - Ranked",
       x = "Diabetes Rate (%)",
       y = "State",
       caption = "Dot plot allows precise comparison of rates") +
  theme_minimal() +
```

```r
  theme(plot.title = element_text(size = 14, face = "bold"),
        axis.text.y = element_text(size = 11))

# Save
ggsave("diabetes_dotplot.png", width = 8, height = 5, dpi = 300)
```

Alternative - Bar Chart:

```r
ggplot(state_health, aes(x = reorder(state, diabetes_rate),
                          y = diabetes_rate, fill = diabetes_rate)) +
  geom_col() +
  coord_flip() +
  scale_fill_gradient(low = "lightblue", high = "darkred") +
  labs(title = "Diabetes Rates by State",
       x = "State", y = "Diabetes Rate (%)") +
  theme_minimal() +
  theme(legend.position = "none")
```

Why this adds value beyond the map: - **Precise values:** Easier to read exact percentages than from color gradients - **Clear ranking:** Immediately shows which states have highest/lowest rates - **Better comparison:** Easier to judge magnitude of differences between states - **Accessibility:** Works better for colorblind viewers - **Complement not duplicate:** Map shows spatial patterns; dot plot shows precise values and ranking

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c) Geographic vs Non-geographic Representations (3 points):

Strengths of Geographic Maps: - Show spatial patterns and clustering - Reveal regional influences (climate, culture, policies) - Help identify border effects or diffusion patterns - Useful for understanding geographic disparities - Good for public communication and awareness

Limitations of Geographic Maps: - Harder to read precise values - Large states dominate visually (area bias) - Color interpretation can be subjective - Not ideal for exact comparisons

Strengths of Non-geographic Visualizations (bar/dot plots): - Precise value reading - Easy magnitude comparisons - Clear ranking of states - Better for detailed analysis - No geographic bias

Limitations: - Lose spatial context - Can't see regional patterns - Miss geographic relationships

When to Use Each:

Use Geographic Maps when: - Spatial patterns are important (disease spread, environmental factors) - Communicating to general public - Showing regional disparities for policy - Investigating neighbor effects or diffusion

Use Non-geographic Charts when: - Precise comparisons needed - Ranking states is the goal - Analyzing correlations with other variables - Presenting to technical audiences - Space is limited (publications)

Best Practice: Use both together for comprehensive analysis - map for spatial context, chart for precise values.

Bonus Question (5 points)

Question 16: Critical Thinking About AI in Biostatistics

Scenario: Researcher wants to use AI to analyze EHR data to identify diabetes risk factors.

a) Two ways AI could be helpful (2 points):

1. **Pattern Recognition in Large Datasets:**
   - AI can process thousands of variables and identify complex interactions that humans might miss
   - Machine learning can detect non-linear relationships between risk factors
   - Example: AI might discover unexpected combinations of medications that correlate with diabetes risk
2. **Rapid Literature Review and Hypothesis Generation:**
   - AI can quickly summarize existing research on diabetes risk factors
   - Can suggest variables to include based on latest evidence
   - Helps researcher avoid reinventing the wheel or missing important confounders

Additional helpful uses: - Natural language processing to extract diagnoses from clinical notes - Automated data cleaning and outlier detection - Generating R/Python code for standard analyses

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b) Two potential problems or limitations (2 points):

1. **Context and Causation Confusion:**
   - AI may identify spurious correlations without understanding biological plausibility
   - May not distinguish between causal risk factors and mere associations
   - Example: AI might suggest "hospital visits" as risk factor when it's actually a consequence
2. **Bias in Training Data and Assumptions:**
   - AI trained on biased EHR data may perpetuate healthcare disparities
   - May not check statistical assumptions (linearity, independence, etc.)
   - Could use outdated diabetes diagnostic criteria if training data is old
   - May miss important confounders or effect modifiers

Additional concerns: - Privacy and ethical issues with patient data - Inability to handle missing data appropriately - Black box nature makes interpretation difficult - May not account for temporal relationships properly

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c) Essential human oversight (1 point):

Critical oversight needed:

1. **Clinical/Domain Expertise:**
   - Biostatistician or epidemiologist must evaluate biological plausibility of findings
   - Clinician should verify that identified risk factors make medical sense
2. **Statistical Validation:**
   - Check all statistical assumptions
   - Validate results with traditional methods
   - Assess for confounding and effect modification
3. **Ethical Review:**
   - Ensure findings don't perpetuate health disparities
   - Verify patient privacy protection
   - Consider equity implications of recommendations
4. **Iterative Refinement:**
   - Human reviews AI output and refines the analysis
   - Tests alternative models and sensitivity analyses
   - Validates findings in independent datasets

Key Principle: AI should augment human expertise, not replace it. The researcher must understand every step of the analysis and take responsibility for the conclusions.

Grading Rubric Summary

Multiple Choice (40 points): 4 points each, auto-graded

Short Answer Questions (30 points): - Question 11 (Data Types): 2 points per variable (correct type + justification) - Question 12 (Geographic): 3 + 4 + 3 points for parts a, b, c - Question 13 (Multi-panel): 3 + 4 + 3 points for parts a, b, c

R Programming (30 points): - Question 14: 5 + 5 + 5 points for parts a, b, c - Code correctness: 70% - Proper output: 20% - Interpretation: 10% - Question 15: 8 + 4 + 3 points for parts a, b, c - Working code: 70% - Visualization quality: 20% - Interpretation: 10%

Bonus (5 points): 2 + 2 + 1 points for thoughtful, evidence-based responses

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Total: 103 points (100 base + 3 bonus)